



Mid-term Exam Review

Presented by Can LI

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Chapter 11



Chapter 11: Infinite Sequences Series

Key Points:

11.4 Comparison Test;

11.5 The Ratio and Root Tests;

11.6 Alternating Series, Absolute and Conditional Convergence;

11.7 Power Series;

11.8 Taylor and Maclaurin Series.

Chapter 11



11.4 Comparison Test

THEOREM 10 The Comparison Test

Let $\sum a_n$ be a series with no negative terms.

- (a) $\sum a_n$ converges if there is a convergent series $\sum c_n$ with $a_n \leq c_n$ for all $n > N$, for some integer N .
- (b) $\sum a_n$ diverges if there is a divergent series of nonnegative terms $\sum d_n$ with $a_n \geq d_n$ for all $n > N$, for some integer N .

THEOREM 11 Limit Comparison Test

Suppose that $a_n > 0$ and $b_n > 0$ for all $n \geq N$ (N an integer).

1. If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c > 0$, then $\sum a_n$ and $\sum b_n$ both converge or both diverge.
2. If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ and $\sum b_n$ converges, then $\sum a_n$ converges.
3. If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

p-series?

Chapter 11



11.5 The Ratio and Root Tests

THEOREM 12 The Ratio Test

Let $\sum a_n$ be a series with positive terms and suppose that

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \rho.$$

Then

- (a) the series *converges* if $\rho < 1$,
- (b) the series *diverges* if $\rho > 1$ or ρ is infinite,
- (c) the test is *inconclusive* if $\rho = 1$.

THEOREM 13 The Root Test

Let $\sum a_n$ be a series with $a_n \geq 0$ for $n \geq N$, and suppose that

$$\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \rho.$$

Then

- (a) the series *converges* if $\rho < 1$,
- (b) the series *diverges* if $\rho > 1$ or ρ is infinite,
- (c) the test is *inconclusive* if $\rho = 1$.

Chapter 11



11.6 Alternating Series, Absolute and Conditional Convergence;

THEOREM 14 The Alternating Series Test (Leibniz's Theorem)

The series

$$\sum_{n=1}^{\infty} (-1)^{n+1} u_n = u_1 - u_2 + u_3 - u_4 + \cdots$$

converges if all three of the following conditions are satisfied:

1. The u_n 's are all positive.
2. $u_n \geq u_{n+1}$ for all $n \geq N$, for some integer N .
3. $u_n \rightarrow 0$.

DEFINITION Absolutely Convergent

A series $\sum a_n$ **converges absolutely** (is **absolutely convergent**) if the corresponding series of absolute values, $\sum |a_n|$, converges.

DEFINITION Conditionally Convergent

A series that converges but does not converge absolutely **converges conditionally**.

Chapter 11



11.6 Alternating Series, Absolute and Conditional Convergence;

1. **The n th-Term Test:** Unless $a_n \rightarrow 0$, the series diverges.
2. **Geometric series:** $\sum ar^n$ converges if $|r| < 1$; otherwise it diverges.
3. **p -series:** $\sum 1/n^p$ converges if $p > 1$; otherwise it diverges.
4. **Series with nonnegative terms:** Try the Integral Test, Ratio Test, or Root Test. Try comparing to a known series with the Comparison Test.
5. **Series with some negative terms:** Does $\sum |a_n|$ converge? If yes, so does $\sum a_n$, since absolute convergence implies convergence.
6. **Alternating series:** $\sum a_n$ converges if the series satisfies the conditions of the Alternating Series Test.

Chapter 11



11.7 Power series

DEFINITIONS Power Series, Center, Coefficients

A **power series about $x = 0$** is a series of the form

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + \cdots + c_n x^n + \cdots \quad (1)$$

A **power series about $x = a$** is a series of the form

$$\sum_{n=0}^{\infty} c_n (x - a)^n = c_0 + c_1 (x - a) + c_2 (x - a)^2 + \cdots + c_n (x - a)^n + \cdots \quad (2)$$

in which the **center** a and the **coefficients** $c_0, c_1, c_2, \dots, c_n, \dots$ are constants.

THEOREM 18 The Convergence Theorem for Power Series

If the power series $\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots$ converges for $x = c \neq 0$, then it converges absolutely for all x with $|x| < |c|$. If the series diverges for $x = d$, then it diverges for all x with $|x| > |d|$.

Chapter 11



11.7 Power series

COROLLARY TO THEOREM 18

The convergence of the series $\sum c_n(x - a)^n$ is described by one of the following three possibilities:

1. There is a positive number R such that the series diverges for x with $|x - a| > R$ but converges absolutely for x with $|x - a| < R$. The series may or may not converge at either of the endpoints $x = a - R$ and $x = a + R$.
2. The series converges absolutely for every x ($R = \infty$).
3. The series converges at $x = a$ and diverges elsewhere ($R = 0$).

How to Test a Power Series for Convergence

1. Use the Ratio Test (or n th-Root Test) to find the interval where the series converges absolutely. Ordinarily, this is an open interval

$$|x - a| < R \quad \text{or} \quad a - R < x < a + R.$$

2. If the interval of absolute convergence is finite, test for convergence or divergence at each endpoint, as in Examples 3a and b. Use a Comparison Test, the Integral Test, or the Alternating Series Test.
3. If the interval of absolute convergence is $a - R < x < a + R$, the series diverges for $|x - a| > R$ (it does not even converge conditionally), because the n th term does not approach zero for those values of x .

Chapter 11



11.8 Taylor and Maclaurin Series

DEFINITIONS Taylor Series, Maclaurin Series

Let f be a function with derivatives of all orders throughout some interval containing a as an interior point. Then the **Taylor series generated by f at $x = a$** is

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x - a)^k = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!} (x - a)^2 + \cdots + \frac{f^{(n)}(a)}{n!} (x - a)^n + \cdots.$$

The **Maclaurin series generated by f** is

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k = f(0) + f'(0)x + \frac{f''(0)}{2!} x^2 + \cdots + \frac{f^{(n)}(0)}{n!} x^n + \cdots,$$

the Taylor series generated by f at $x = 0$.

Chapter 11



11.8 Taylor and Maclaurin Series

TABLE 11.1 Frequently used Taylor series

$$\frac{1}{1-x} = 1 + x + x^2 + \cdots + x^n + \cdots = \sum_{n=0}^{\infty} x^n, \quad |x| < 1$$

$$\frac{1}{1+x} = 1 - x + x^2 - \cdots + (-x)^n + \cdots = \sum_{n=0}^{\infty} (-1)^n x^n, \quad |x| < 1$$

$$e^x = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad |x| < \infty$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad |x| < \infty$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + (-1)^n \frac{x^{2n}}{(2n)!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \quad |x| < \infty$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots + (-1)^{n-1} \frac{x^n}{n} + \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n}, \quad -1 < x \leq 1$$

$$\ln \frac{1+x}{1-x} = 2 \tanh^{-1} x = 2 \left(x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots + \frac{x^{2n+1}}{2n+1} + \cdots \right) = 2 \sum_{n=0}^{\infty} \frac{x^{2n+1}}{2n+1}, \quad |x| < 1$$

$$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots + (-1)^n \frac{x^{2n+1}}{2n+1} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}, \quad |x| \leq 1$$

Binomial Series

$$\begin{aligned} (1+x)^m &= 1 + mx + \frac{m(m-1)x^2}{2!} + \frac{m(m-1)(m-2)x^3}{3!} + \cdots + \frac{m(m-1)(m-2)\cdots(m-k+1)x^k}{k!} + \cdots \\ &= 1 + \sum_{k=1}^{\infty} \binom{m}{k} x^k, \quad |x| < 1, \end{aligned}$$

where

$$\binom{m}{1} = m, \quad \binom{m}{2} = \frac{m(m-1)}{2!}, \quad \binom{m}{k} = \frac{m(m-1)\cdots(m-k+1)}{k!} \quad \text{for } k \geq 3.$$

Note: To write the binomial series compactly, it is customary to define $\binom{m}{0}$ to be 1 and to take $x^0 = 1$ (even in the usually excluded case where $x = 0$), yielding $(1+x)^m = \sum_{k=0}^{\infty} \binom{m}{k} x^k$. If m is a *positive integer*, the series terminates at x^m and the result converges for all x .

Chapter 12&13



Chapter 12: Vectors and the Geometry of Space

Key Points:

12.3 The Dot Product;

12.4 The Cross Product;

12.5 Lines and Planes in Space.

Chapter 13: Vector-Valued Functions and Motion in Space

Key Points:

13.3 Arc Length and the Unit Tangent Vector T ;

13.4 Curvature and the Unit Normal Vector N ;

13.5 Torsion and the Unit Binormal Vector B .

Chapter 12&13



12.3 The Dot Product

THEOREM 1 Angle Between Two Vectors

The angle θ between two nonzero vectors $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ is given by

$$\theta = \cos^{-1} \left(\frac{u_1 v_1 + u_2 v_2 + u_3 v_3}{|\mathbf{u}| |\mathbf{v}|} \right).$$

DEFINITION Dot Product

The **dot product** $\mathbf{u} \cdot \mathbf{v}$ (“**u dot v**”) of vectors $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ is

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + u_3 v_3.$$

DEFINITION Orthogonal Vectors

Vectors $\mathbf{u}$ and $\mathbf{v}$ are **orthogonal** (or **perpendicular**) if and only if $\mathbf{u} \cdot \mathbf{v} = 0$.

Properties of the Dot Product

If $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are any vectors and c is a scalar, then

1. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
2. $(c\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (c\mathbf{v}) = c(\mathbf{u} \cdot \mathbf{v})$
3. $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$
4. $\mathbf{u} \cdot \mathbf{u} = |\mathbf{u}|^2$
5. $\mathbf{0} \cdot \mathbf{u} = 0$.

Vector projection of $\mathbf{u}$ onto $\mathbf{v}$:

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2} \right) \mathbf{v} \quad (1)$$

Scalar component of $\mathbf{u}$ in the direction of $\mathbf{v}$:

$$|\mathbf{u}| \cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|} = \mathbf{u} \cdot \frac{\mathbf{v}}{|\mathbf{v}|} \quad (2)$$

Chapter 12&13



12.4 The Cross Product

DEFINITION Cross Product

$$\mathbf{u} \times \mathbf{v} = (|\mathbf{u}| |\mathbf{v}| \sin \theta) \mathbf{n}$$

Properties of the Cross Product

If $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are any vectors and r, s are scalars, then

1. $(r\mathbf{u}) \times (s\mathbf{v}) = (rs)(\mathbf{u} \times \mathbf{v})$
2. $\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = \mathbf{u} \times \mathbf{v} + \mathbf{u} \times \mathbf{w}$
3. $(\mathbf{v} + \mathbf{w}) \times \mathbf{u} = \mathbf{v} \times \mathbf{u} + \mathbf{w} \times \mathbf{u}$
4. $\mathbf{v} \times \mathbf{u} = -(\mathbf{u} \times \mathbf{v})$
5. $\mathbf{0} \times \mathbf{u} = \mathbf{0}$

$$|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}| |\mathbf{v}| |\sin \theta| |\mathbf{n}| = |\mathbf{u}| |\mathbf{v}| \sin \theta.$$

Calculating Cross Products Using Determinants

If $\mathbf{u} = u_1 \mathbf{i} + u_2 \mathbf{j} + u_3 \mathbf{k}$ and $\mathbf{v} = v_1 \mathbf{i} + v_2 \mathbf{j} + v_3 \mathbf{k}$, then

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}.$$

Chapter 12&13



12.5 Lines and Planes in space

Parametric Equations for a Line

The standard parametrization of the line through $P_0(x_0, y_0, z_0)$ parallel to $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$ is

$$x = x_0 + tv_1, \quad y = y_0 + tv_2, \quad z = z_0 + tv_3, \quad -\infty < t < \infty \quad (3)$$

Distance from a Point S to a Line Through P Parallel to $\mathbf{v}$

$$d = \frac{|\vec{PS} \times \mathbf{v}|}{|\mathbf{v}|} \quad (5)$$

Equation for a Plane

The plane through $P_0(x_0, y_0, z_0)$ normal to $\mathbf{n} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$ has

Vector equation: $\mathbf{n} \cdot \vec{P_0P} = 0$

Component equation: $A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$

Component equation simplified: $Ax + By + Cz = D$, where
 $D = Ax_0 + By_0 + Cz_0$

Chapter 12&13



Q1. Suppose $|\vec{a}| = 2$, $|\vec{b}| = 5$, and the angle $\langle \vec{a}, \vec{b} \rangle = \frac{\pi}{3}$. Please find the value of

$$(\vec{a} + 2\vec{b}) \cdot (\vec{a} - 3\vec{b}).$$

Solution:

$$\begin{aligned}(\vec{a} + 2\vec{b}) \cdot (\vec{a} - 3\vec{b}) &= |\vec{a}|^2 - 6|\vec{b}|^2 - \vec{a} \cdot \vec{b} = |\vec{a}|^2 - 6|\vec{b}|^2 - |\vec{a}| |\vec{b}| \cos\theta \\ &= -146 - 5\sqrt{3}\end{aligned}$$

Q2. Find the vector of projection of $\vec{a} = \langle 1, -3, 4 \rangle$ onto $\vec{b} = \langle 2, 2, 1 \rangle$, and the scalar component of $\vec{a}$ in the direction of $\vec{b}$.

Solution:

The projection vector of $\vec{a}$ in the direction of $\vec{b}$ is:

$$\text{proj}_{\vec{b}} \vec{a} = \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} \right) \vec{b} = \vec{0}$$

So that the scalar of projection is 0.

Chapter 12&13



Q3. Suppose the two sides of a parallelogram are two vectors $\vec{a} = \langle 2, -3, 1 \rangle$ and $\vec{b} = \langle 2, -1, 3 \rangle$. Find the area of the parallelogram.

Solution:

$$\text{Area} = |\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin \theta = 4\sqrt{6}$$

Q4. Find an equation for the plane through $(1, -2, 3)$ parallel to the plane $3x + 2y - z = 5$.

Solution:

The normal vector of the plane is $n = \langle 3, 2, -1 \rangle$.

So that the plane is:

$$3(x - 1) + 2(y + 2) - (z - 3) = 0$$

Simplify the structure:

$$3x + 2y - z = -4$$

Chapter 12&13



Chapter 13 Key points

Formulas for Curves in Space

Unit tangent vector: $\mathbf{T} = \frac{\mathbf{v}}{|\mathbf{v}|}$

Principal unit normal vector: $\mathbf{N} = \frac{d\mathbf{T}/dt}{|d\mathbf{T}/dt|}$

Binormal vector: $\mathbf{B} = \mathbf{T} \times \mathbf{N}$

Curvature: $\kappa = \left| \frac{d\mathbf{T}}{ds} \right| = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3}$

Torsion: $\tau = -\frac{d\mathbf{B}}{ds} \cdot \mathbf{N} = \frac{\begin{vmatrix} \dot{x} & \dot{y} & \dot{z} \\ \ddot{x} & \ddot{y} & \ddot{z} \\ \ddot{x} & \ddot{y} & \ddot{z} \end{vmatrix}}{|\mathbf{v} \times \mathbf{a}|^2}$

Tangential and normal scalar components of acceleration: $\mathbf{a} = a_T \mathbf{T} + a_N \mathbf{N}$

$$a_T = \frac{d}{dt} |\mathbf{v}|$$

$$a_N = \kappa |\mathbf{v}|^2 = \sqrt{|\mathbf{a}|^2 - a_T^2}$$

Chapter 12&13



(a) (3 marks) Find the arc length parametrization of the helix $\vec{r}(t) = \langle a \cos t, a \sin t, bt \rangle$,

where a and b are constants and $a > 0$.

(b) (7 marks) Based on the arc length parametrization of the helix, find the three Frenet

frame vectors $\vec{T}, \vec{N},$ and $\vec{B}$, the curvature and torsion for the helix.

Solution:

(a) $\vec{r}(t) = \langle a \cos t, a \sin t, bt \rangle$,

$$v(t) = \langle -a \sin t, a \cos t, b \rangle, |v(t)| = \sqrt{a^2 + b^2} \text{ -----1 mark}$$

$$s(t) = \int_{t_0}^t |v(t)| dt = \sqrt{a^2 + b^2} (t - t_0) \text{ -----1 mark}$$

$$t_0 = 0 \text{ s, } s(t) = \sqrt{a^2 + b^2} t \text{ -----1 mark}$$

(b)

$$T = \frac{v}{|v|} = \left\langle \frac{-a \sin t}{\sqrt{a^2 + b^2}}, \frac{a \cos t}{\sqrt{a^2 + b^2}}, b \right\rangle \text{ -----1 mark}$$

$$N = \frac{\frac{dT}{dt}}{\left| \frac{dT}{dt} \right|} = \langle -\cos t, -\sin t, 0 \rangle \text{ -----2 marks}$$

$$B = T \times N = \left\langle \frac{b \sin t}{\sqrt{a^2 + b^2}}, \frac{-b \cos t}{\sqrt{a^2 + b^2}}, \frac{a}{\sqrt{a^2 + b^2}} \right\rangle \text{ -----2 marks}$$

$$\kappa = \left| \frac{dT}{ds} \right| = \frac{a}{a^2 + b^2} \text{ -----1 mark}$$

$$\tau = -\frac{dB}{ds} \cdot N = \frac{b}{a^2 + b^2} \text{ -----1 mark}$$

Chapter 14



Chapter 14: Partial Derivatives

Key points:

14.2 Limits and Continuity in Higher Dimensions;

14.4 The Chain Rule;

14.5 Directional Derivatives and Gradient Vectors;

14.6 Tangent Planes and Differentials;

14.7 Extreme Values and Saddle Points;

14.8 Lagrange Multipliers.

Chapter 14



14.2 Limits and Continuity in Higher Dimensions;

DEFINITION Continuous Function of Two Variables

A function $f(x, y)$ is **continuous at the point** (x_0, y_0) if

1. f is defined at (x_0, y_0) ,
2. $\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y)$ exists,
3. $\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = f(x_0, y_0)$.

A function is **continuous** if it is continuous at every point of its domain.

Two-Path Test for Nonexistence of a Limit

If a function $f(x, y)$ has different limits along two different paths as (x, y) approaches (x_0, y_0) , then $\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y)$ does not exist.

Show that

$$f(x, y) = \begin{cases} \frac{2xy}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

is continuous at every point except the origin (Figure 14.11).

Show that the function

$$f(x, y) = \frac{2x^2y}{x^4 + y^2}$$

(Figure 14.12) has no limit as (x, y) approaches $(0, 0)$.

Chapter 14



14.3 Partial Derivatives

DEFINITION Partial Derivative with Respect to x

The **partial derivative of $f(x, y)$ with respect to x** at the point (x_0, y_0) is

$$\left. \frac{\partial f}{\partial x} \right|_{(x_0, y_0)} = \lim_{h \rightarrow 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h},$$

provided the limit exists.

DEFINITION Partial Derivative with Respect to y

The **partial derivative of $f(x, y)$ with respect to y** at the point (x_0, y_0) is

$$\left. \frac{\partial f}{\partial y} \right|_{(x_0, y_0)} = \left. \frac{d}{dy} f(x_0, y) \right|_{y=y_0} = \lim_{h \rightarrow 0} \frac{f(x_0, y_0 + h) - f(x_0, y_0)}{h},$$

provided the limit exists.

THEOREM 2 The Mixed Derivative Theorem

If $f(x, y)$ and its partial derivatives f_x , f_y , f_{xy} , and f_{yx} are defined throughout an open region containing a point (a, b) and are all continuous at (a, b) , then

$$f_{xy}(a, b) = f_{yx}(a, b).$$

Chapter 14



14.3 Partial Derivatives

COROLLARY OF THEOREM 3 **Continuity of Partial Derivatives Implies Differentiability**

If the partial derivatives f_x and f_y of a function $f(x, y)$ are continuous throughout an open region R , then f is differentiable at every point of R .

THEOREM 4 **Differentiability Implies Continuity**

If a function $f(x, y)$ is differentiable at (x_0, y_0) , then f is continuous at (x_0, y_0) .

Chapter 14



18. Continuous extension Let

$$f(x, y) = \begin{cases} \frac{\sin(x - y)}{|x| + |y|}, & |x| + |y| \neq 0 \\ 0, & (x, y) = (0, 0). \end{cases}$$

Is f continuous at the origin? Why?

Differentiable?

The partial derivatives of x and y exist?

Chapter 14



14.4 The Chain Rule

THEOREM 5 Chain Rule for Functions of Two Independent Variables

If $w = f(x, y)$ has continuous partial derivatives f_x and f_y and if $x = x(t)$, $y = y(t)$ are differentiable functions of t , then the composite $w = f(x(t), y(t))$ is a differentiable function of t and

$$\frac{df}{dt} = f_x(x(t), y(t)) \cdot x'(t) + f_y(x(t), y(t)) \cdot y'(t),$$

or

$$\frac{dw}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}.$$

THEOREM 6 Chain Rule for Functions of Three Independent Variables

If $w = f(x, y, z)$ is differentiable and x , y , and z are differentiable functions of t , then w is a differentiable function of t and

$$\frac{dw}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt}.$$

Chapter 14



14.4 The Chain Rule

THEOREM 7 Chain Rule for Two Independent Variables and Three Intermediate Variables

Suppose that $w = f(x, y, z)$, $x = g(r, s)$, $y = h(r, s)$, and $z = k(r, s)$. If all four functions are differentiable, then w has partial derivatives with respect to r and s , given by the formulas

DEFINITION Directional Derivative

The **derivative of f at $P_0(x_0, y_0)$ in the direction of the unit vector $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$** is the number

$$\left(\frac{df}{ds}\right)_{\mathbf{u}, P_0} = \lim_{s \rightarrow 0} \frac{f(x_0 + su_1, y_0 + su_2) - f(x_0, y_0)}{s}, \quad (1)$$

provided the limit exists.

THEOREM 8 A Formula for Implicit Differentiation

Suppose that $F(x, y)$ is differentiable and that the equation $F(x, y) = 0$ defines y as a differentiable function of x . Then at any point where $F_y \neq 0$,

$$\frac{dy}{dx} = -\frac{F_x}{F_y}.$$

Chapter 14



设 $z = xy + xF(u)$, $u = \frac{y}{x}$, $F(u)$ 为可导函数, 证明: $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = z + xy$

Chapter 14



14.5 Directional Derivatives and Gradient Vectors

DEFINITION Directional Derivative

The derivative of f at $P_0(x_0, y_0)$ in the direction of the unit vector $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$ is the number

$$\left(\frac{df}{ds}\right)_{\mathbf{u}, P_0} = \lim_{s \rightarrow 0} \frac{f(x_0 + su_1, y_0 + su_2) - f(x_0, y_0)}{s}, \quad (1)$$

provided the limit exists.

DEFINITION Gradient Vector

The gradient vector (gradient) of $f(x, y)$ at a point $P_0(x_0, y_0)$ is the vector

$$\nabla f = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j}$$

obtained by evaluating the partial derivatives of f at P_0 .

THEOREM 9 The Directional Derivative Is a Dot Product

If $f(x, y)$ is differentiable in an open region containing $P_0(x_0, y_0)$, then

$$\left(\frac{df}{ds}\right)_{\mathbf{u}, P_0} = (\nabla f)_{P_0} \cdot \mathbf{u}, \quad (4)$$

the dot product of the gradient f at P_0 and $\mathbf{u}$.

Properties of the Directional Derivative $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} = |\nabla f| \cos \theta$

1. The function f increases most rapidly when $\cos \theta = 1$ or when $\mathbf{u}$ is the direction of ∇f . That is, at each point P in its domain, f increases most rapidly in the direction of the gradient vector ∇f at P . The derivative in this direction is

$$D_{\mathbf{u}}f = |\nabla f| \cos(0) = |\nabla f|.$$

2. Similarly, f decreases most rapidly in the direction of $-\nabla f$. The derivative in this direction is $D_{\mathbf{u}}f = |\nabla f| \cos(\pi) = -|\nabla f|$.
3. Any direction $\mathbf{u}$ orthogonal to a gradient $\nabla f \neq 0$ is a direction of zero change in f because θ then equals $\pi/2$ and

$$D_{\mathbf{u}}f = |\nabla f| \cos(\pi/2) = |\nabla f| \cdot 0 = 0.$$

Chapter 14



14.6 Tangent Plane and Differentials

DEFINITIONS Tangent Plane, Normal Line

The **tangent plane** at the point $P_0(x_0, y_0, z_0)$ on the level surface $f(x, y, z) = c$ of a differentiable function f is the plane through P_0 normal to $\nabla f|_{P_0}$.

The **normal line** of the surface at P_0 is the line through P_0 parallel to $\nabla f|_{P_0}$.

Plane Tangent to a Surface $z = f(x, y)$ at $(x_0, y_0, f(x_0, y_0))$

The plane tangent to the surface $z = f(x, y)$ of a differentiable function f at the point $P_0(x_0, y_0, z_0) = (x_0, y_0, f(x_0, y_0))$ is

$$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) - (z - z_0) = 0. \quad (4)$$

DEFINITIONS Linearization, Standard Linear Approximation

The **linearization** of a function $f(x, y)$ at a point (x_0, y_0) where f is differentiable is the function

$$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0). \quad (5)$$

The approximation

$$f(x, y) \approx L(x, y)$$

is the **standard linear approximation** of f at (x_0, y_0) .

DEFINITION Total Differential

If we move from (x_0, y_0) to a point $(x_0 + dx, y_0 + dy)$ nearby, the resulting change

$$df = f_x(x_0, y_0) dx + f_y(x_0, y_0) dy$$

in the linearization of f is called the **total differential** of f .

Chapter 14

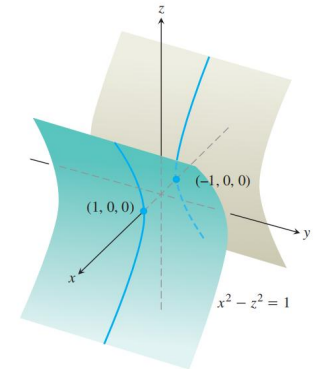


14.7 Extreme Values and Saddle Points

THEOREM 11 Second Derivative Test for Local Extreme Values

Suppose that $f(x, y)$ and its first and second partial derivatives are continuous throughout a disk centered at (a, b) and that $f_x(a, b) = f_y(a, b) = 0$. Then

- i. f has a **local maximum** at (a, b) if $f_{xx} < 0$ and $f_{xx}f_{yy} - f_{xy}^2 > 0$ at (a, b) .
- ii. f has a **local minimum** at (a, b) if $f_{xx} > 0$ and $f_{xx}f_{yy} - f_{xy}^2 > 0$ at (a, b) .
- iii. f has a **saddle point** at (a, b) if $f_{xx}f_{yy} - f_{xy}^2 < 0$ at (a, b) .
- iv. **The test is inconclusive** at (a, b) if $f_{xx}f_{yy} - f_{xy}^2 = 0$ at (a, b) . In this case, we must find some other way to determine the behavior of f at (a, b) .



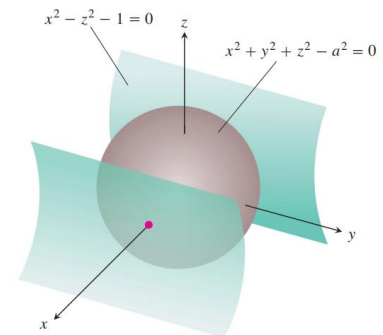
14.8 Lagrange Multipliers

The Method of Lagrange Multipliers

Suppose that $f(x, y, z)$ and $g(x, y, z)$ are differentiable. To find the local maximum and minimum values of f subject to the constraint $g(x, y, z) = 0$, find the values of x, y, z , and λ that simultaneously satisfy the equations

$$\nabla f = \lambda \nabla g \quad \text{and} \quad g(x, y, z) = 0. \quad (1)$$

For functions of two independent variables, the condition is similar, but without the variable z .



Chapter 14



Suppose that f has continuous second partial derivatives, and find the second-order partial derivatives of function $z = f(\sin x, \cos y, e^{x+y})$

$$\frac{\partial^2 z}{\partial x^2}, \frac{\partial^2 z}{\partial x \partial y}, \frac{\partial^2 z}{\partial y^2}$$

Find the extreme values and saddle points of $f(x, y) = x^3 + y^3 + 3xy + 1$.

Mock Exam



Compliance

I have read and understood the rules, regulations and associated penalties detailed in the NPU Student Handbook. I hereby agree to comply with these rules and regulations and uphold the integrity of the university's examination process.

signature :

Number: _____

Mid-term-Examination Paper of NPU

2021-2022 school year, 2nd semester

Teaching school Queen Mary University Of London Engineering

School, NPU ; Lesson **Advanced Math II** Class hours **88**

Date of examination **May 6th, 2022** Exam time **2** hours

Form of exam (closed) (A) volume.

Grade	
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Part 1: MCQs & Fill in the blanks



I. Multiple Selection Exercises.(In one exercise, only one selection is correct. 24 marks in total, 4 marks for per question.)

1. Set $a_n \leq c_n \leq b_n$, and series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are convergent, so series $\sum_{n=1}^{\infty} c_n$ ().

A. Converges.

B. Diverges.

C. May not be convergent, but $\lim_{n \rightarrow +\infty} c_n = 0$. D. $\lim_{n \rightarrow +\infty} c_n$ exists but may not equal to 0.

2. The power series $\sum_{n=1}^{\infty} a_n (x-1)^n$ converges conditionally at $x=-2$, so at $x=2$, the series ().

A. Converges absolutely.

B. Converges conditionally.

C. Diverges.

D. A, B, C are wrong.

Part 1: MCQs & Fill in the blanks

3. Which of the following statements are **not true** if $f(x,y)$ is differentiable at (x_0, y_0) ?

- A. If $\mathbf{u}$ is a unit vector, the derivative of f at (x_0, y_0) in the direction of $\mathbf{u}$ is $(f_x((x_0, y_0))\mathbf{i} + f_y(x_0, y_0)\mathbf{j}) \cdot \mathbf{u}$.
- B. The derivative of f at (x_0, y_0) in the direction of $\mathbf{u}$ is a vector.
- C. The directional derivative of f at (x_0, y_0) has its greatest value in the direction of ∇f .
- D. At (x_0, y_0) , vector of ∇f is normal to the curve $f(x, y) = f(x_0, y_0)$.

4. The value of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{2^n n}$ is ().

- A. $\ln \frac{2}{3}$
- B. $\ln \frac{3}{2}$
- C. $e^{\frac{2}{3}}$
- D. $e^{\frac{3}{2}}$

5. The convergence radius of series $\sum_{n=1}^{\infty} \frac{1}{2^n} \left(1 - \frac{n^n}{n!}\right) x^n$ is ().

- A. $\frac{e}{2}$
- B. $\frac{2}{e}$
- C. $\sqrt{e}$
- D. $\frac{\sqrt{e}}{e}$

6. The angle between $x+y=1$ and $2x+y-2z=2$ is ().

- A. $\frac{\pi}{6}$
- B. $\frac{\pi}{3}$
- C. $\frac{\pi}{4}$
- D. $\frac{\pi}{2}$

Part 2: Comprehensive Questions



II. Comprehensive Questions. (You must write down some steps necessarily.)

7. (8 marks) For the series:

$$\sum_{n=1}^{\infty} (-1)^n \frac{\ln(n)}{n}$$

(a) Does the series converge absolutely, converges conditionally, or diverge? (just write down the result of your justification, 2 marks)

(b) Prove your conclusion determined in part (a). (6 marks)

Part 2: Comprehensive Questions



8. (8 marks) Find the radius and interval of convergence of series as follows:

$$\sum_{n=1}^{\infty} \frac{(x-1)^n}{n 2^n}$$

Part 2: Comprehensive Questions



9. (10 marks)

(a) Find the convergence interval of the series:

$$\sum_{n=0}^{\infty} (n+1)x^n$$

And, within the interval, find the sum of the series as a function of x . (6 marks)

(b) According to the result in (a), try to find the value of

$$\sum_{n=1}^{\infty} \frac{n}{3^{n-1}}$$

(4 marks)

Part 2: Comprehensive Questions



10. (8 marks) Find the Taylor series of function $f(x) = \frac{1}{x^2}$ at $x = 2$.

Part 2: Comprehensive Questions



11. (8 marks)

(a)(6 marks) The graph $y = f(x)$ in the xy -plane automatically has the parametrization form. Use this formula to show that if f is a twice-differentiable function of x , then:

$$\kappa(x) = \frac{|f''(x)|}{(1 + [f'(x)]^2)^{\frac{3}{2}}}$$

(b)(2 marks) Use the formula in question (a), find the curvature of $y = \sin x$.

Part 2: Comprehensive Questions



12. (12 marks) Given a function $f(x, y) = \begin{cases} \frac{x^2 y^2}{x^2 + y^2}, & x^2 + y^2 \neq 0 \\ 0, & (x, y) = (0, 0) \end{cases}$.

Show that:

- (1) the function is continuous at $(0, 0)$;
- (2) the function has two partial derivatives at $(0, 0)$;
- (3) the function is differentiable at $(0, 0)$.

Part 2: Comprehensive Questions



13. (8 marks) Find $\frac{\partial z}{\partial y}$ and $\frac{\partial^2 z}{\partial x \partial y}$ if the equation $z = x^3 F(xy, \frac{y}{x})$ and F has continuous second-order partial derivatives.

Part 2: Comprehensive Questions



14. (6 marks) Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ if the equation $F(x - y, y - z) = 0$ defines z as a function of the two independent variables x and y and F has continuous partial derivatives.

Part 2: Comprehensive Questions



15. (6 marks) Find the extreme values and saddle points of the function:

$$f(x, y) = x^3 + y^3 + 3xy + 1.$$

Summary



- **Questions ????**



Thank you for listening